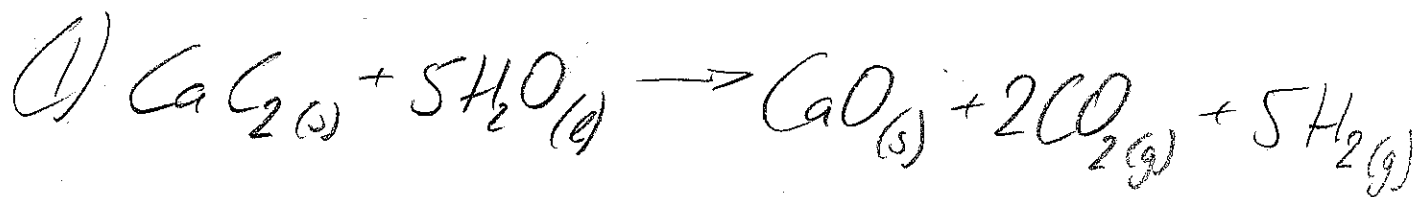


①

Homework #10

SolutionsCHE 210, Fall '08
Yevgenny Strashnov
(TA)

$$\Delta H_r^\circ = (+) 69.36 \text{ kJ/mol} \quad \left(\begin{array}{l} \text{standard heat} \\ \text{of reaction} \end{array} \right)$$

a) The rxn. is endothermic, i.e. it "absorbs" heat energy as it proceeds in the forward direction.

We can tell this from the fact that ΔH_r° is positive.

Therefore, if the rxn is run under adiabatic conditions (no heat exchange w/ surroundings), the reactor temp. would decrease & the reactor would have to be heated in order to maintain the temp. constant. From the perspective of Physical Chemistry, the energy required to break all the bonds between the atoms of the reactants is more than the energy released when the bonds between the atoms of the product species are formed, i.e. there is a net deficit of energy which makes the rxn endothermic...

②

b)

$$\Delta \hat{U}_c^{\circ} = \Delta \hat{H}_c^{\circ} - RT \left[\sum \nu_i - \sum \nu_i \right]$$

gaseous products
gaseous reactants

Stoichiometric coeff.
from the balanced
chemical rxn.

5 from H_2 & 2 from CO_2

$$= 69.36 \text{ kJ/mol} - 8.314 \text{ J/mol}\cdot\text{K} \cdot \frac{4 \text{ J}}{1000 \text{ J}} \cdot 298.15 \text{ K} (7 - 0)$$

no gaseous react.

$$\Delta \hat{U}_c^{\circ} = 52.0 \text{ kJ/mol}$$

note: $\Delta \hat{U}_c^{\circ}$ is the change in internal energy when

1 g-mol (or simply 1 mol) of $CaH_2(s)$ & 5 mol of $H_2O(l)$

all @ 25°C & 1 atm (standard T & P or STP) react
to form 1 mol of $CaO(s)$, 2 mol of $CO_2(g)$ &
5 mol of $H_2(g)$, also all @ STP.

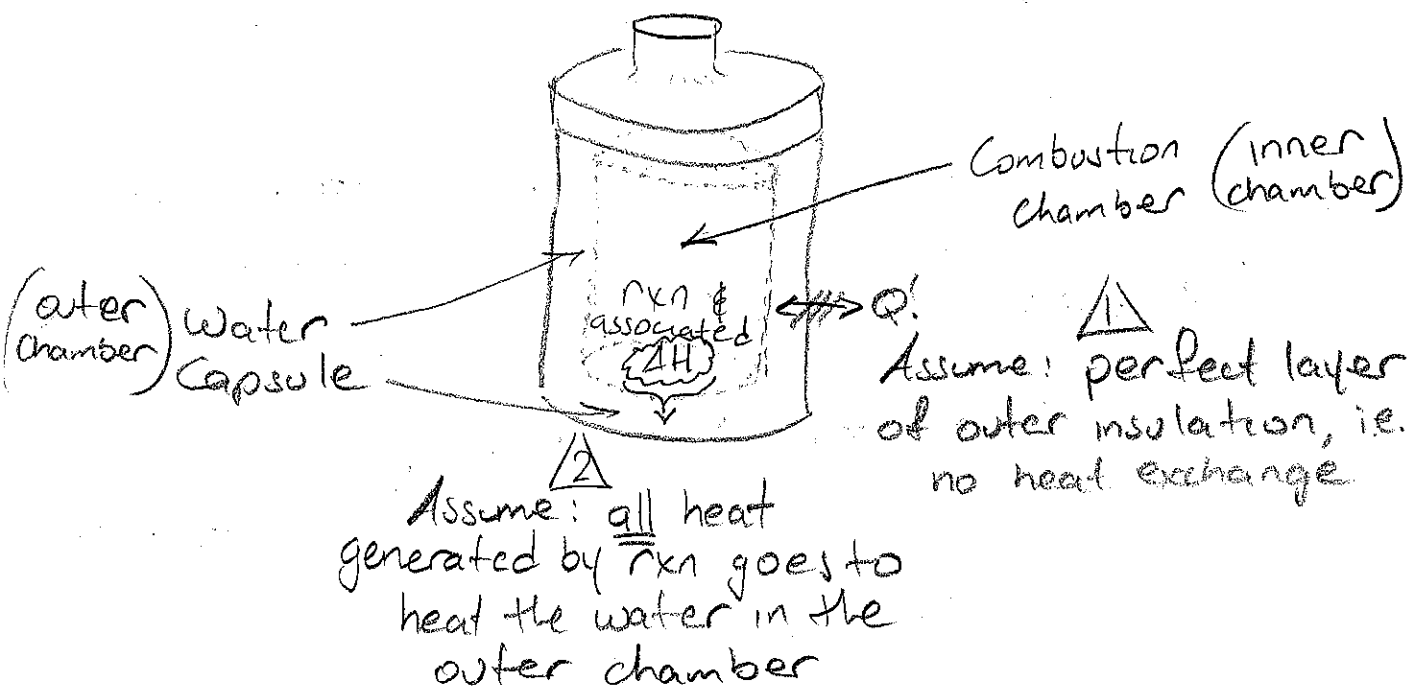
$$c) \quad Q = \Delta U = \frac{n_{CaH_2} \cdot \Delta \hat{U}_c^{\circ}}{\nu_{CaH_2}} = 150 \text{ g } CaH_2 \cdot \frac{\text{mol}}{64.10 \text{ g}} \cdot \frac{52.0 \text{ kJ}}{\text{mol } CaH_2}$$

$$Q = (+) 121.7 \text{ kJ}$$

By convention, the positive
sign in front of the value for Q means that heat
will flow into the reactor & add to the overall
internal energy of the system. This also matches
up w/ our answer to part (a).

(2) Murphy 6.32

Process Diagram:



First, take the water inside the outer chamber as the system...

$$\Rightarrow U_{H_2O, f} - U_{H_2O, i} = Q = \Delta H_{rxn}$$

$$C_v(H_2O, l) = C_p(H_2O, l) \equiv 4.18 \text{ J/g} \cdot ^\circ\text{C}$$

$$\text{Then, } \Delta U_{H_2O} = Q = m_{H_2O} C_{v, H_2O, l} \cdot \Delta T_{H_2O} = 650 \text{ g} \cdot 4.18 \text{ J/g} \cdot ^\circ\text{C} \cdot (36.2 - 23.6) ^\circ\text{C}$$

$$Q = 34,234 \text{ J}$$

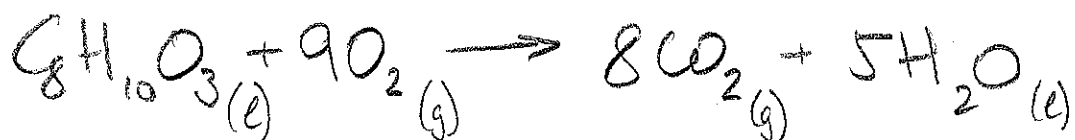
④

Next, take the contents of the combustion chamber as our new system...

From this perspective, the system gives off the energy (heat, $Q = 34.234 \text{ kJ}$) that our first system receives.....

$$\therefore \Delta U_{\text{rxn}} = -Q = -34,234 \text{ J}$$

We proceed to examine the chemical reactions:



According to Assumption 2 (see diagram), we neglect any temp. change inside the chamber. We can also reasonably assume that the rxn. proceeds at or near STP ³ conditions, which allows us to use most supplemental data that's contained within the tables & appendices without much consideration.

Thus, we can write:

$$\Delta U_{\text{rxn}} = n_r \Delta \hat{U}_r = -34,234 \text{ J}$$

5

$$n_r = 2.7g \cdot \frac{\text{mol}}{154g} = .0175 \text{ mol} = \xi_{\text{rxn}}$$

$$\text{Then, } \Delta \hat{U}_r = \frac{-34,234 \text{ J}}{.0175 \text{ mol}} = 1,952.6 \text{ kJ/mol}$$

Next, assume that the unknown compound & the water are in the liquid state. ⁴ We can then calculate the change in the number of moles in the gas phase between the products & the reactants: $8 - 9 = -1$

$$\begin{aligned} \text{Therefore, } \Delta \hat{H}_r &= \Delta \hat{U}_r + RT \Delta n_{r(g)} \\ &= -1952.6 \text{ kJ} + .0083144 \text{ kJ/mol} \cdot \text{K} \end{aligned}$$

$$\Delta \hat{H}_r = -1958 \text{ kJ/mol} \quad .298 \text{ K} \cdot (-1) \text{ mol}$$

Finally, we can write:

$$\Delta \hat{H}_r = \sum \nu_i \Delta \hat{H}_{f,i} = (-1) \Delta \hat{H}_{f, \text{C}_8\text{H}_{10}\text{O}_3} + (-9) \overset{\text{O}_{2(g)}}{(0)} + 8 \overset{\text{CO}_{2(g)}}{(-395.5)}$$

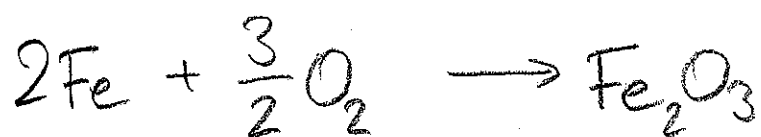
$$\text{Solve for } \Delta \hat{H}_{f, \text{C}_8\text{H}_{10}\text{O}_3} + 5(-285.84) = -1958 \text{ kJ/mol}$$

$$\boxed{\Delta \hat{H}_{f, \text{C}_8\text{H}_{10}\text{O}_3} \cong -2621 \text{ kJ/mol}} \quad \text{H}_2\text{O}(l)$$

⑥

(3) Murphy 6.34

Chemical Reaction:



We can note that this is the elemental formation reaction for Fe_2O_3 , so we can look up $\Delta \hat{H}_f$ directly:

$$\Delta \hat{H}_{f, \text{Fe}_2\text{O}_3} = -830.5 \text{ kJ/mol}$$

Next, we note that we need to produce 100 kJ/hr for 6 hrs, or a net amount of energy equal to

$$100 \text{ kJ/hr} \cdot 6 \text{ hrs} = 600 \text{ kJ}$$

$$\text{Then, } 600 \text{ kJ} \cdot \frac{\text{mol Fe}_2\text{O}_3}{830.5 \text{ kJ}} = \underline{.722 \text{ mol Fe}_2\text{O}_3}$$

We would therefore need to place twice as many moles of pure Fe into the pack initially (according to the rxn. stoichiometry) ...

(7)

$0.722 \cdot 2 = 1.44 \text{ mol Fe}$ need to be in the pack initially

$$1.44 \text{ mol Fe} \cdot \frac{56 \text{ g}}{\text{mol}} \approx \boxed{80 \text{ g Fe}}$$

How much more would the hot pack weigh by the time the reaction ran to completion?

@ $t = 0$, the mass of the pack is $\sim 80 \text{ g}$ (Fe only)

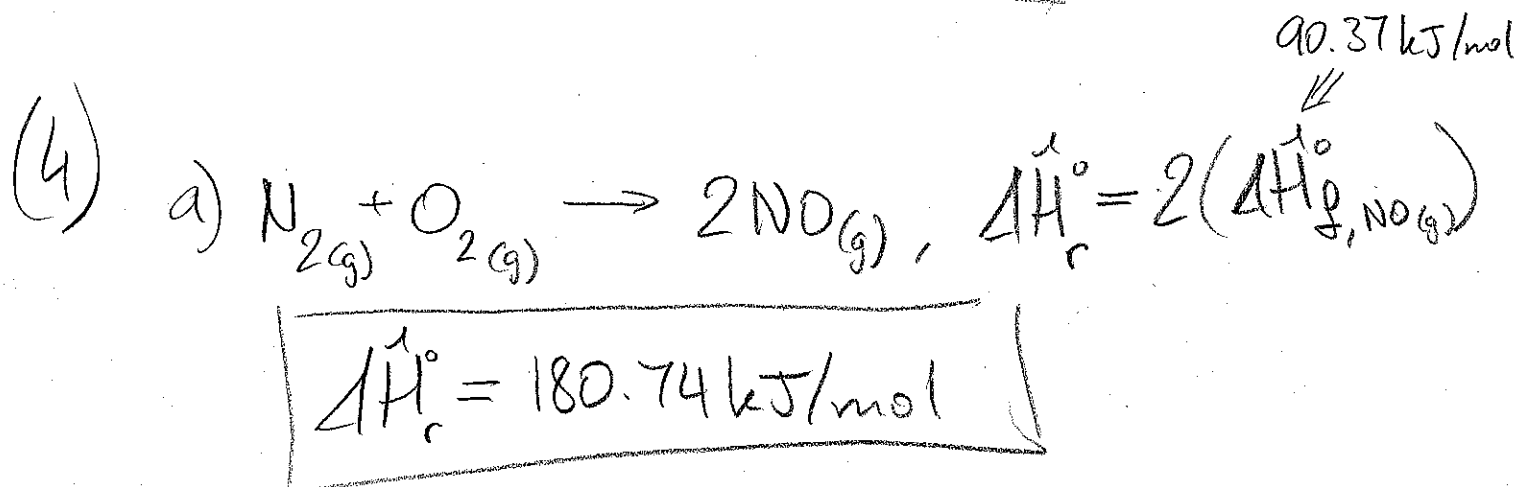
@ $t = t_{\text{comp}}$, the mass of the pack is:

$$0.722 \text{ mol } \underset{\text{Fe}_2\text{O}_3}{\text{Fe}_2\text{O}_3} \cdot \frac{160 \text{ g}}{\underset{\text{Fe}_2\text{O}_3}{\text{mol}}} = 115.22 \text{ g}$$

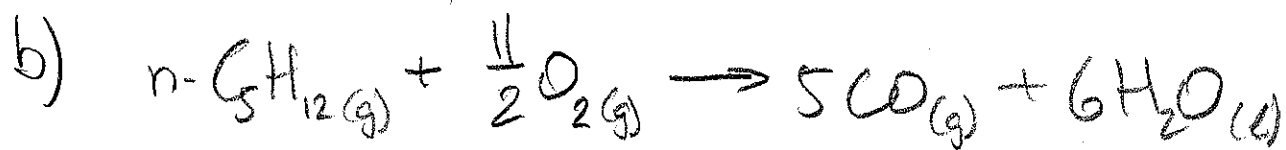
$$\Delta m = 115.22 - 80 \approx \boxed{35.5 \text{ g} = \Delta m}$$

or

$$\frac{35.5}{80} \times 100\% = \boxed{44.38\%}$$



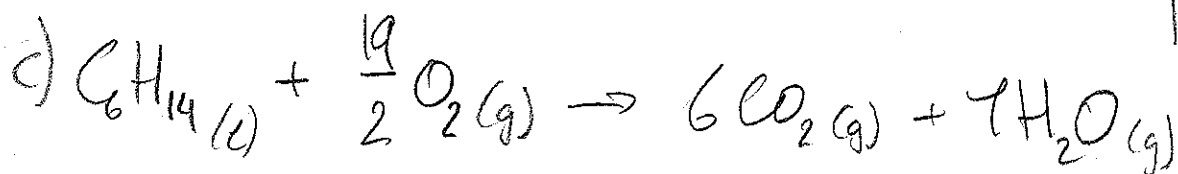
8)



$$\Delta \hat{H}_r^\circ = 5(\Delta \hat{H}_{f, \text{CO}(\text{g})}^\circ) + 6(\Delta \hat{H}_{f, \text{H}_2\text{O}(\text{l})}^\circ)$$

$$- \Delta \hat{H}_{f, \text{n-C}_5\text{H}_{12}}^\circ$$

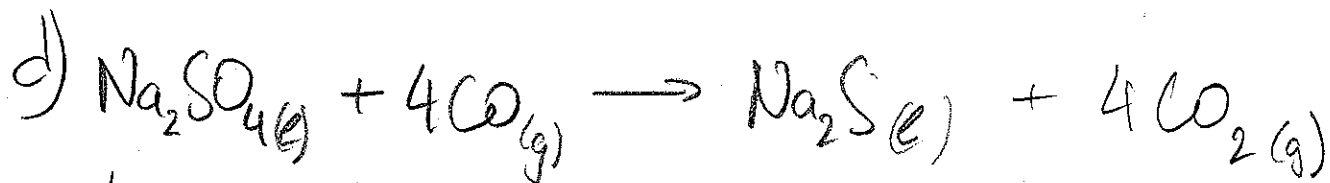
$$= [5(-110.52) + 6(-285.84) - (-146.4)] \text{ kJ/mol} = \boxed{-2121.2 \text{ kJ/mol}} \\ = \Delta \hat{H}_r^\circ$$



$$\Delta \hat{H}_r^\circ = 6(\Delta \hat{H}_{f, \text{CO}_2(\text{g})}^\circ) + 7(\Delta \hat{H}_{f, \text{H}_2\text{O}(\text{g})}^\circ) - (\Delta \hat{H}_{f, \text{C}_6\text{H}_{14}(\text{l})}^\circ)$$

$$= [6(-393.5) + 7(-241.83) - (-73.0)] \text{ kJ/mol}$$

$$= \boxed{-3855 \text{ kJ/mol} = \Delta \hat{H}_r^\circ}$$



$$\Delta \hat{H}_r^\circ = \Delta \hat{H}_{f, \text{Na}_2\text{S}(\text{s})}^\circ + 4(\Delta \hat{H}_{f, \text{CO}_2(\text{g})}^\circ) - 4(\Delta \hat{H}_{f, \text{CO}(\text{g})}^\circ) - \Delta \hat{H}_{f, \text{Na}_2\text{SO}_4(\text{s})}^\circ$$

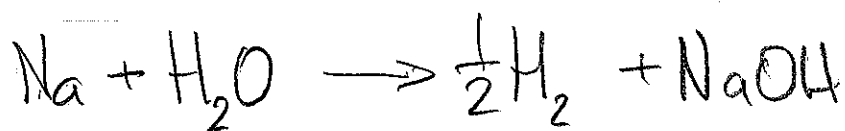
$$= [(-373.2 + 6.7) + 4(-393.5) - 4(-110.52) - (-1384.5 + 24.3)] \text{ kJ/mol}$$

(9)

$$\Delta H_r^\circ = -138.2 \text{ kJ/mol}$$

(5) Murphy 6.37

Chemical Reaction:



$$\begin{aligned}\Delta H_r^\circ &= \sum_i \nu_{i1} \cdot \Delta H_{f,i}^\circ = (-1)(-285.84) + 1(-469.15) \\ &= -183 \text{ kJ/mol}\end{aligned}$$

Since $1 \text{ mL H}_2\text{O} \approx 1 \text{ g H}_2\text{O}$ (around STP),

$$\approx 1 \text{ g} / 18 \text{ g/mol} = .056 \text{ mol H}_2\text{O}$$

$$100 \text{ g Na} \cdot \frac{\text{mol}}{23 \text{ g}} \approx 4.35 \text{ mol Na}$$

$\therefore \text{H}_2\text{O}$ is the limiting reagent...

Once the reaction runs to completion, the

(10)

$4.35 - .056 = 4.29$ mol Na remain unreacted...

Heat capacity of solid Na = $28 \text{ J/mol} \cdot ^\circ\text{C}$

So, we have:

$$\sum_{\text{rxn}} \Delta H_r^\circ = .056 \text{ mol} \cdot 183000 \text{ J/mol} =$$

$$= M_{\text{Na}} \cdot C_{\text{P Na}} \cdot \Delta T_{\text{Na}} = 4.29 \text{ mol} \cdot 28 \text{ J/mol} \cdot ^\circ\text{C} \cdot \Delta T_{\text{Na}}$$

$$\therefore \Delta T_{\text{Na}} = \frac{.056 \cdot 183 \times 10^3}{4.29 \cdot 28} = \boxed{85.31^\circ\text{C} = \Delta T_{\text{Na}}}$$

It becomes exceedingly clear that once all of the 1 ml of H_2O has reacted, the heat evolved from the reaction will make the Na metal very hot!

If we also recall that the reaction produces

H_2 (which is highly flammable in the presence of air or O_2), we can deduce why it may be very dangerous to splash water onto solid Alkali metals (grp I) such as Na: the H_2 might ~~EXPLODE~~ 